

Database Technologies – Assignment 5

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
w1_99099_vaibhavi>SELECT CONCAT(LOWER(first_name), '-' , UPPER(last_name)) as EMPLOYEE_FULL_NAME FROM employees;
+-----+
| EMPLOYEE_FULL_NAME |
+-----+
| ellen-ABEL         |
| sundar-ANDE        |
| mozhe-ATKINSON     |
| shelli-BAIDA       |
| amit-BANDA         |
| elizabeth-BATES    |
| sarah-BELL         |
| david-BERNSTEIN     |
| laura-BISSOT        |
| harrison-BLOOM     |
| hermann-BROWN      |
| alexis-BULL        |
| anthony-CABRIO     |
| gerald-CAMBRAULT   |
| nanette-CAMBRAULT  |
| john-CHEN          |
| kelly-CHUNG        |
| karen-COLMENARES   |
| curtis-DAVIES      |
| pat-DAVIS          |
| julia-DELLINGER    |
| jennifer-DILLY     |
| louise-DORAN       |
| alberto-ERRAZURIZ  |
| britney-EVERETT    |
| daniel-FAVIET      |
| kevin-FEENEY       |
| jean-FLEAUR        |
| taylor-FOX         |
| adam-FRIPP         |
| lex-GARCIA         |
| ki-GEE             |
```

```
donald-CONNELL
christopher-OLSEN
tj-OLSON
lisa-OZER
karen-PARTNERS
joshua-PATEL
randall-PERKINS
hazel-PHILTANKER
luis-POPP
trenna-RAJS
michael-ROGERS
nandita-SARCHAND
ismael-SCIARRA
john-SEO
sarath-SEWALL
john-SINGH
lindsey-SMITH
william-SMITH
stephen-STILES
martha-SULLIVAN
patrick-SULLY
jonathon-TAYLOR
winston-TAYLOR
sigal-TOBIAS
sean-TUCKER
oliver-TUVAULT
jose manuel-URMAN
peter-VARGAS
timothy-VENZL
clara-VISHNEY
shanta-VOLLMAN
alana-WALSH
matthew-WEISS
jennifer-WHALEN
david-WILLIAMS
neena-YANG
eleni-ZLOTKEY
+-----+
107 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

2. Display the first word of the job title. (Hint : Use substring_index).

```
w1_99099_vaibhavi>SELECT SUBSTRING_INDEX(job_title, ' ', 1) AS first_word_of_title
-> FROM jobs;
+-----+
| first_word_of_title |
+-----+
| Public              |
| Accounting          |
| Administration      |
| President           |
| Administration      |
| Accountant          |
| Finance             |
| Human               |
| Programmer          |
| Marketing            |
| Marketing            |
| Public              |
| Purchasing          |
| Purchasing          |
| Sales               |
| Sales               |
| Shipping            |
| Stock               |
| Stock               |
+-----+
19 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
w1_99099_vaibhavi>SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME)AS FIRST_NAME_LENGTH FROM EMPLOYEES
-> WHERE SUBSTRING(last_name, 4) LIKE '%b%';
+-----+-----+-----+
| FIRST_NAME | LAST_NAME | FIRST_NAME_LENGTH |
+-----+-----+-----+
| Gerald     | Cambrault | 6                  |
| Nanette    | Cambrault | 7                  |
| Nancy      | Gruenberg | 5                  |
| Susan      | Jacobs    | 5                  |
+-----+-----+-----+
4 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

```
w1_99099_vaibhavi>SELECT employee_id, first_name, last_name FROM employees
-> where first_name LIKE '% %'
-> or last_name LIKE '% %';
```

employee_id	first_name	last_name
112	Jose Manuel	Urman

```
1 row in set (0.00 sec)

w1_99099_vaibhavi>
```

5. Display first name, salary, and round the salary to thousands.

```
w1_99099_vaibhavi> SELECT first_name, salary, round(salary, -3) FROM employees;
```

first_name	salary	round(salary, -3)
Steven	24000.00	24000
Neena	17000.00	17000
Lex	17000.00	17000
Alexander	9000.00	9000
Bruce	6000.00	6000
David	4800.00	5000
Valli	4800.00	5000
Diana	4200.00	4000
Nancy	12008.00	12000
Daniel	9000.00	9000
John	8200.00	8000
Ismael	7700.00	8000
Jose Manuel	7800.00	8000
Luis	6900.00	7000
Den	11000.00	11000
Alexander	3100.00	3000
Shelli	2900.00	3000
Sigal	2800.00	3000
Guy	2600.00	3000
Karen	2500.00	3000
Matthew	8000.00	8000
Adam	8200.00	8000
Payam	7900.00	8000
Shanta	6500.00	7000

Jonathan	8800.00	9000
Jack	8400.00	8000
Kimberely	7000.00	7000
Charles	6200.00	6000
Winston	3200.00	3000
Jean	3100.00	3000
Martha	2500.00	3000
Girard	2800.00	3000
Nandita	4200.00	4000
Alexis	4100.00	4000
Julia	3400.00	3000
Anthony	3000.00	3000
Kelly	3800.00	4000
Jennifer	3600.00	4000
Timothy	2900.00	3000
Randall	2500.00	3000
Sarah	4000.00	4000
Britney	3900.00	4000
Samuel	3200.00	3000
Vance	2800.00	3000
Alana	3100.00	3000
Kevin	3000.00	3000
Donald	2600.00	3000
Douglas	2600.00	3000
Jennifer	4400.00	4000
Michael	13000.00	13000
Pat	6000.00	6000
Susan	6500.00	7000
Hermann	10000.00	10000
Shelley	12008.00	12000
William	8300.00	8000

+-----+-----+

107 rows in set (0.00 sec)

w1_99099_vaibhavi>

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

```
w1_99099_vaibhavi>SELECT employee_id FROM employees e
w1_99099_vaibhavi>SELECT employee_id, DATE_FORMAT(end_date, '%W, %M %D %Y')
AS formatted_end_date
-> FROM job_history;
```

employee_id	formatted_end_date
101	Thursday, October 27th 2011
101	Sunday, March 15th 2015
102	Sunday, July 24th 2016
114	Sunday, December 31st 2017
122	Sunday, December 31st 2017
176	Saturday, December 31st 2016
176	Sunday, December 31st 2017
200	Friday, June 17th 2011
200	Saturday, December 31st 2016
201	Tuesday, December 19th 2017

10 rows in set (0.01 sec)

```
w1_99099_vaibhavi>
```

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST_DAY).

```
w1_99099_vaibhavi>SELECT first_name,  
->      ADDDATE(LAST_DAY(SUBDATE(hire_date, INTERVAL 1 MONTH)), INTERV  
AL 1 DAY) AS first_salary_date  
-> FROM employees;
```

+-----+-----+	
first_name	first_salary_date
+-----+-----+	
Steven	2013-06-01
Neena	2015-09-01
Lex	2011-01-01
Alexander	2016-01-01
Bruce	2017-05-01
David	2015-06-01
Valli	2016-02-01
Diana	2017-02-01
Nancy	2012-08-01
Daniel	2012-08-01
John	2015-09-01
Ismael	2015-09-01
Jose Manuel	2016-03-01
Luis	2017-12-01
Den	2012-12-01
Alexander	2013-05-01
Shelli	2015-12-01
Sigal	2015-07-01
Guy	2016-11-01
Karen	2017-08-01
Matthew	2014-07-01
Adam	2015-04-01
Payam	2013-05-01
Shanta	2015-10-01
Kevin	2017-11-01
Julia	2015-07-01
Irene	2016-09-01
James	2017-01-01

Kimberely	2017-05-01
Charles	2018-01-01
Winston	2016-01-01
Jean	2016-02-01
Martha	2017-06-01
Girard	2018-02-01
Nandita	2014-01-01
Alexis	2015-02-01
Julia	2016-06-01
Anthony	2017-02-01
Kelly	2015-06-01
Jennifer	2015-08-01
Timothy	2016-07-01
Randall	2017-12-01
Sarah	2014-02-01
Britney	2015-03-01
Samuel	2016-07-01
Vance	2017-03-01
Alana	2016-04-01
Kevin	2016-05-01
Donald	2017-06-01
Douglas	2018-01-01
Jennifer	2013-09-01
Michael	2014-02-01
Pat	2015-08-01
Susan	2012-06-01
Hermann	2012-06-01
Shelley	2012-06-01
William	2012-06-01

+-----+

107 rows in set (0.01 sec)

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
w1_99099_vaibhavi>SELECT first_name,
->      CONCAT(
->          TIMESTAMPDIFF(YEAR, hire_date, CURDATE()), ' Years, ',
->          TIMESTAMPDIFF(MONTH, hire_date, CURDATE()) % 12, ' Months,
',
->          TIMESTAMPDIFF(DAY, hire_date + INTERVAL TIMESTAMPDIFF(MONT
H, hire_date, CURDATE()) MONTH, CURDATE()), ' Days'
->      ) AS exact_experience
-> FROM employees;
```

first_name	exact_experience
Steven	13 Years, 2 Months, 23 Days
Neena	10 Years, 11 Months, 19 Days
Lex	15 Years, 7 Months, 27 Days
Alexander	10 Years, 8 Months, 6 Days
Bruce	9 Years, 3 Months, 19 Days
David	11 Years, 2 Months, 15 Days
Valli	10 Years, 7 Months, 4 Days
Diana	9 Years, 7 Months, 2 Days
Nancy	14 Years, 0 Months, 23 Days
Daniel	14 Years, 0 Months, 24 Days
John	10 Years, 11 Months, 12 Days
Ismael	10 Years, 11 Months, 10 Days
Jose Manuel	10 Years, 6 Months, 2 Days
Luis	8 Years, 9 Months, 2 Days
Den	13 Years, 9 Months, 2 Days
Alexander	13 Years, 3 Months, 22 Days
Shelli	10 Years, 8 Months, 16 Days
Sigal	11 Years, 1 Months, 16 Days
Guy	9 Years, 9 Months, 25 Days
Karen	9 Years, 0 Months, 30 Days
Matthew	12 Years, 1 Months, 22 Days
Adam	11 Years, 4 Months, 30 Days
Payam	13 Years, 4 Months, 8 Days
Shanta	10 Years, 10 Months, 30 Days
Kevin	8 Years, 9 Months, 24 Days
Julia	11 Years, 1 Months, 24 Days
Irene	9 Years, 11 Months, 12 Days
James	9 Years, 7 Months, 26 Days

```
| Alyssa | 11 Years, 5 Months, 21 Days |
| Jonathon | 10 Years, 5 Months, 16 Days |
| Jack | 10 Years, 4 Months, 17 Days |
| Kimberely | 9 Years, 3 Months, 16 Days |
| Charles | 8 Years, 8 Months, 5 Days |
| Winston | 10 Years, 7 Months, 16 Days |
| Jean | 10 Years, 6 Months, 17 Days |
| Martha | 9 Years, 2 Months, 19 Days |
| Girard | 8 Years, 7 Months, 6 Days |
| Nandita | 12 Years, 7 Months, 13 Days |
| Alexis | 11 Years, 6 Months, 20 Days |
| Julia | 10 Years, 2 Months, 16 Days |
| Anthony | 9 Years, 7 Months, 2 Days |
| Kelly | 11 Years, 2 Months, 26 Days |
| Jennifer | 11 Years, 0 Months, 27 Days |
| Timothy | 10 Years, 1 Months, 29 Days |
| Randall | 8 Years, 8 Months, 21 Days |
| Sarah | 12 Years, 7 Months, 5 Days |
| Britney | 11 Years, 6 Months, 6 Days |
| Samuel | 10 Years, 2 Months, 8 Days |
| Vance | 9 Years, 5 Months, 23 Days |
| Alana | 10 Years, 4 Months, 16 Days |
| Kevin | 10 Years, 3 Months, 17 Days |
| Donald | 9 Years, 2 Months, 19 Days |
| Douglas | 8 Years, 7 Months, 27 Days |
| Jennifer | 12 Years, 11 Months, 23 Days |
| Michael | 12 Years, 6 Months, 23 Days |
| Pat | 11 Years, 0 Months, 23 Days |
| Susan | 14 Years, 3 Months, 2 Days |
| Hermann | 14 Years, 3 Months, 2 Days |
| Shelley | 14 Years, 3 Months, 2 Days |
| William | 14 Years, 3 Months, 2 Days |
+-----+
107 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
w1_99099_vaibhavi>SELECT employee_id, first_name, hire_date AS joining_date
FROM employees
-> WHERE MONTHNAME(hire_date) = 'August';
```

employee_id	first_name	joining_date
108	Nancy	2012-08-17
109	Daniel	2012-08-16
119	Karen	2017-08-10
129	Laura	2015-08-20
134	Michael	2016-08-26
152	Peter	2015-08-20
158	Allan	2014-08-01
189	Jennifer	2015-08-13
202	Pat	2015-08-17

```
9 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

10. Display the details of employees who joined before the year 2015.

```
w1_99099_vaibhavi>SELECT * FROM employees
-> WHERE year(hire_date) < 2015;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR	12000.00	NULL	101	100
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	8000.00	NULL	100	50
122	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	7900.00	NULL	100	50
133	Jason	Mallin	JMALLIN	1.650.555.0133	2014-06-14	ST_CLERK	3300.00	NULL	122	50
137	Renske	Ladwig	RLADWIG	1.650.555.0137	2013-07-14	ST_CLERK	3600.00	NULL	123	50
141	Trenna	Rajs	TRAJS	1.650.555.0141	2013-10-17	ST_CLERK	3500.00	NULL	124	50
145	John	Singh	JSINGH	44.1632.960000	2014-10-01	SA_MAN	14000.00	0.40	100	80
156	Janette	King	JKING	44.1632.960011	2014-01-30	SA_REP	10000.00	0.35	146	80
157	Patrick	Sully	PSULLY	44.1632.960012	2014-03-04	SA_REP	9500.00	0.35	146	80
158	Allan	McEwen	AMCEWEN	44.1632.960013	2014-08-01	SA_REP	9000.00	0.35	146	80
174	Ellen	Abel	EABEL	44.1632.960029	2014-05-11	SA_REP	11000.00	0.30	149	80
184	Nandita	Sarchand	NSARCHAN	1.650.555.0149	2014-01-27	SH_CLERK	4200.00	NULL	121	50
192	Sarah	Bell	SBELL	1.650.555.0157	2014-02-04	SH_CLERK	4000.00	NULL	123	50
200	Jennifer	Whalen	JWHALEN	1.515.555.0165	2013-09-17	AD_ASST	4000.00	NULL	101	10
201	Michael	Martinez	MMARTINE	1.515.555.0166	2014-02-17	MK_MAN	13000.00	NULL	100	20
203	Susan	Jacobs	SJACOBS	1.515.555.0168	2012-06-07	HR_REP	6500.00	NULL	101	40
204	Hermann	Brown	HBROWN	1.515.555.0169	2012-06-07	PR_REP	10000.00	NULL	101	70
205	Shelley	Higgins	SHIGGINS	1.515.555.0170	2012-06-07	AC_MGR	12000.00	NULL	101	110
206	William	Gietz	WGIEZT	1.515.555.0171	2012-06-07	AC_ACCOUNT	8300.00	NULL	205	110

```
24 rows in set (0.00 sec)

w1_99099_vaibhavi>
```

11. Display the number of days between system date and 1st January 2011.

```
w1_99099_vaibhavi>SELECT DATEDIFF(CURDATE(), '2011-01-01') AS days_passed;
```

days_passed
5730

```
1 row in set (0.03 sec)

w1_99099_vaibhavi>
```

12. Display number of employees joined after 15th of any month.

```
w1_99099_vaibhavi>SELECT COUNT(*) AS total_employees
-> FROM employees
-> WHERE DAY(hire_date) > 15;
+-----+
| total_employees |
+-----+
|          57 |
+-----+
1 row in set (0.00 sec)

w1_99099_vaibhavi>
```

13. Display third highest salary of employees

```
r 'LIMIT 2,1' at line 5
w1_99099_vaibhavi>SELECT DISTINCT salary
-> FROM employees
-> ORDER BY salary DESC
-> LIMIT 2, 1;
+-----+
| salary |
+-----+
| 14000.00 |
+-----+
1 row in set (0.00 sec)

w1_99099_vaibhavi>
```